

3.4.1 Manufacturing Cells

- Product family or group technology departments aggregate medium volume-variety parts into families based on similar manufacturing operations or design attributes. The machines that are required to manufacture the part family are grouped together to form a “cell”.
- Manufacturing cells group machines, employees, materials, tooling, and material handling and storage equipment to produce family of parts.
- The most important benefits of cellular manufacturing are achieved when manufacturing cells are designed, controlled, and operated using *Just-In-Time (JIT)*,

- **Direct** clustering algorithm (DCA) is a methodology for identification of cellular manufacturing structure within an existing manufacturing shop
- The steps of this methodology is
 1. Order the rows and columns. Sum the 1s in each column & row. Order of rows (top to bottom) is descending order of the number of 1s in each where ties exist, break the ties in descending order numerical sequence.
 2. Sort the C. Beginning with the first row of matrix, shift to the left of the matrix all columns having a 1 in the first row. Cont. the process row by row until no further opportunities exist for shifting columns.
 3. Sort the Rows. Column by column, beginning with the leftmost column, shift rows until upward when opportunities exists to form blocks of 1s. Performing the column and row sortation is facilitated by using spreadsheets.
 4. From cells look for opportunities to form cells such that all processing for each part occurs in a single cell.

The below matrix is a representation of the machines visited by a set of parts during production. Our objective is to group parts and machines together to form cells.

		M/C #																											
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26		
Part #	1	1		1		1			1	1		1						1	1				1						9
	2	1		1		1			1	1		1						1	1				1						9
	3	1		1		1			1	1		1						1	1				1						9
	4	1		1		1			1	1		1						1	1				1						9
	5	1		1			1		1	1		1						1	1				1						9
	6	1		1			1		1	1		1						1	1				1						9
	7	1		1			1		1	1		1						1	1				1						9
	8		1		1			1				1		1							1	1				1			8
	9		1		1			1				1		1							1	1				1			8
	10	1		1											1	1	1	1					1			1	1	1	10
	11	1		1											1	1	1	1					1			1	1	1	10
	12	1		1											1	1	1	1					1			1	1	1	10
	13	1		1											1	1	1	1					1			1	1	1	10
		11	2	11	2	4	3	2	7	7	2	7	2	4	4	4	4	7	7	2	2	4	7	2	4	4	4		

The DCA methodology accomplishes this by ranking each row and column by its number of occurrences, that is “1”s. This ranking is represented by the number of far right of each row and the bottom of each column. Once the ranking is performed, the rows and columns are to be sorted in descending order

	Machine #																										
Part #	23	20	19	12	10	7	4	2	6	26	25	24	21	16	15	14	13	5	22	18	21	17	11	9	3	1	# of ls
13										1	1	1	1	1	1	1	1								1	1	10
12										1	1	1	1	1	1	1	1								1	1	10
11										1	1	1	1	1	1	1	1								1	1	10
10										1	1	1	1	1	1	1	1								1	1	10
7									1										1	1	1	1	1	1	1	1	9
6									1										1	1	1	1	1	1	1	1	9
5									1										1	1	1	1	1	1	1	1	9
4																		1	1	1	1	1	1	1	1	1	9
3																		1	1	1	1	1	1	1	1	1	9
2																		1	1	1	1	1	1	1	1	1	9
1																		1	1	1	1	1	1	1	1	1	9
9	1	1	1	1	1	1	1	1																			8
8	1	1	1	1	1	1	1	1																			8
# of ls	2	2	2	2	2	2	2	2	3	4	4	4	4	4	4	4	4	4	7	7	7	7	7	7	11	11	

The next step in the methodology is to; starting with the first column, transfer all rows with occurrences to the top of the matrix.

[illegible]

	Machine #																												
Part #	26	25	24	21	16	15	14	13	3a	1a	22	18	17	11	9	8	3b	1b	6	5	23	20	19	12	10	7	4	2	
13	1	1	1	1	1	1	1	1	1	1																			
12	1	1	1	1	1	1	1	1	1	1																			
11	1	1	1	1	1	1	1	1	1	1																			
10	1	1	1	1	1	1	1	1	1	1																			
7	Cell A										1	1	1	1	1	1	1	1	1	1									
6											1	1	1	1	1	1	1	1	1	1									
5											1	1	1	1	1	1	1	1	1	1									
4											1	1	1	1	1	1	1	1	1		1								
3											1	1	1	1	1	1	1	1	1		1								
2											1	1	1	1	1	1	1	1	1		1								
1											1	1	1	1	1	1	1	1	1		1		Cell C						
9											Cell B										1	1	1	1	1	1	1	1	1
8																					1	1	1	1	1	1	1	1	1

3.4.2 Layout Type base upon Material Flow